TreeDiff: AST-Guided Code Generation with Diffusion LLMs

Syntax-aware masking and loss weighting for structured code denoising

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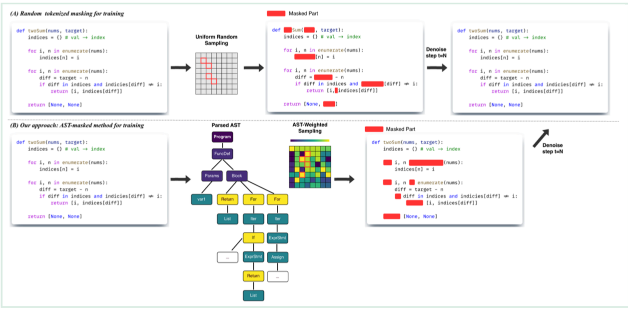
Motivation

- Autoregressive code LLMs (LLaDA) decode **entire sequences token-by-token**
- Diffs contain **three distinct regions** (prompt / reasoning / code) with different masking semantics
- **Flat span-level masking ignores this structure** — destroys reasoning chains while preserving redundant prompt tokens

Method: Tree-Guided Diffusion



TreeDiff augments masked diffusion with **AST-aware policies**: the syntax tree guides which tokens to mask and how to weight the denoising loss, preserving syntactic structure throughout the diffusion process.

Region-Specific Masking

Each diff region gets a tailored masking policy:

Prompt **intact**
Reasoning **random**
Code **AST-weighted**

AST Node Weighting

Loss weights scale with AST node depth: **deeper nodes (constructors, conditionals) receive higher weight** than surface tokens.

```
w(node) = 1 + alpha * depth(node)
```

Node Type	Depth	Weight
FunctionDef	1	1.5
arguments	2	2.0
If / For	3	2.5
Compare / Call	4+	3.0+

Main Results

+13.3%
Relative improvement on HumanEval+

42.1
Pass@1 score (50 denoising steps)

Method	HEval	HEval+	MBPP	Steps
LLaDA (base)	36.6	33.5	39.2	256
LLaDA + random mask	37.2	34.1	40.0	256
LLaDA + span mask	38.4	35.6	41.3	256
TreeDiff (ours)	**43.9**	**42.1**	**46.8**	**50**

TreeDiff achieves higher scores with 5x fewer denoising steps.

Ablation Study

Configuration	HEval+	Delta
Random masking (baseline)	34.1	--
+ AST-weighted loss	37.8	+3.7
+ Region-specific masking	39.4	+5.3
Full TreeDiff	**42.1**	**+8.0**

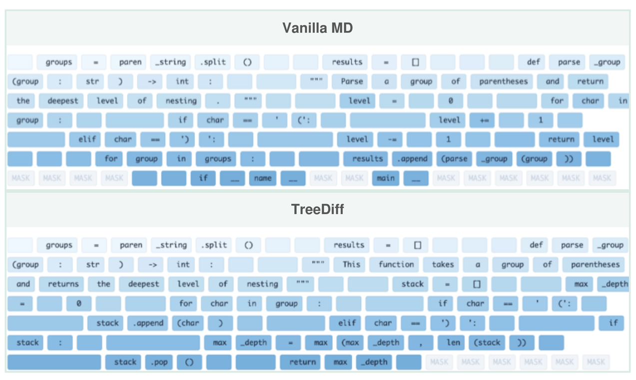
> **Key insight:** Random masking outperforms span masking on code diffs. Spans often split syntactic units (e.g., an if-condition), while random masking preserves more complete AST nodes.

Robustness to Sequence Length

Method	T=512	T=1024
LLaDA + span mask	35.6	34.2
TreeDiff	**42.1**	**41.8**

TreeDiff maintains performance as sequence length doubles, while span masking degrades due to fragmented syntax trees.

Generation Trajectory



Edit distance across denoising steps. Vanilla MD oscillates (54 → 52 → 41 → 46 → 32). TreeDiff converges monotonically with fewer steps.

Only 50 denoising steps (vs 256 decode steps) needed

Qualitative Example

Vanilla MD (Random Mask)

```
count = 0
for char in group:
  if char == '(':
    count += 1
  # Fails to track nested
  # peak depth (no stack)
```

TreeDiff (AST-guided)

```
for char in group:
  if char == '(':
    stack.append(char)
    max_depth = max(
      max_depth, len(stack))
  elif char == ')':
    if stack: stack.pop()
```

HumanEval/6: Maximum nesting depth of parentheses

Key Contributions & Conclusion

- First work to systematically study masking strategies for diffusing code diffs at scale
- Novel AST-aware masking and loss weighting that amplifies structural signal
- Achieve **+13.3%** relative HumanEval+ and **42.1** Pass@1 with only 50 denoising steps — lower FLOPs than 256-step autoregressive decoding
- Practical recipe for adapting pretrained masked diffusion LMs to structured sequence-to-sequence tasks without catastrophic forgetting

We show that the diffusion process on code diffs is not just a noise-to-text mapping, but a structured reconstruction guided by syntax trees.

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GitHub: `Tempest2281/treeDiff-paper`
Code + Weights + Data

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